Signal Chp-2 Math

2.4

Given system -> $x(t) = u(t)$; $h(t) = e^{-at}u(t)$

a) Compute the output $y(t)$ by equation

$$y(t) = x(t) * h(t)$$

b) Compute the output $y(t)$ by equation

$$y(t) = h(t) * x(t)$$



Ans

a) Given,

$$y(t) = x(t) * h(t)$$

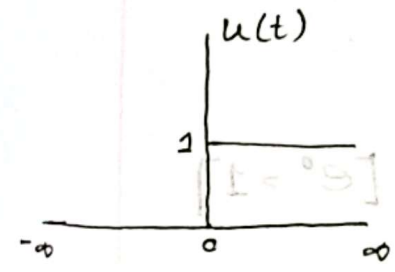
$$\begin{cases} h(t) = e^{-at}u(t) \\ h(k-t) = e^{-a(k+t)}u(k-t) \end{cases}$$

$$= \int_{-\infty}^{\infty} x(t) \cdot h(k-t)\, dt$$

$$= \int_{-\infty}^{\infty} u(t) \cdot e^{-a(k-t)} u(k-t)\, dt$$

$$= \int_{-\infty}^{0} u(t) \cdot e^{-a(k-t)} u(k-t)\, dt + \int_{0}^{\infty} u(t) \cdot e^{-a(k-t)} u(k-t)\, dt$$

Unit step function

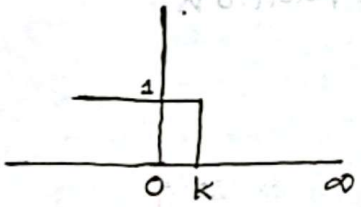


$$= \int_{-\infty}^{0} 0 \cdot e^{-a(k-t)} \cdot u(k-t)\, dt + \int_{0}^{\infty} 1 \cdot e^{-a(k-t)} \cdot u(k-t)\, dt$$

$$= 0 + \int_{0}^{\infty} e^{-a(k-t)} u(k-t)\, dt$$

$$= \int_{0}^{k} e^{-a(k-t)} u(k-t)\, dt + \int_{k}^{\infty} e^{-a(k-t)} u(k-t)\, dt$$

$u(k-t) = -u(t-k)$



$$= \int_{0}^{k} e^{-ak+at} \cdot 1\, dt + \int_{k}^{\infty} e^{-a(k-t)} \cdot 0\, dt$$

$$= \int_{0}^{k} e^{-ak} \cdot e^{at}\, dt$$

$$= e^{-ak} \int_{0}^{k} e^{at}\, dt$$

$$= e^{-ak} \left[\frac{e^{at}}{a} \right]_{0}^{k}$$

$$= e^{-ak} \left[\frac{e^{ak}}{a} - \frac{e^{0}}{a} \right]$$

$$= \frac{e^{-ak} \cdot e^{ak}}{a} - \frac{e^{-ak}}{a} \qquad [e^{0} = 1]$$

$$= \frac{e^{-ak+ak}}{a} - \frac{e^{-ak}}{a}$$

$$= \frac{1}{a} - \frac{e^{-ak}}{a}$$

$$= \frac{1}{a}\left(1 - e^{-ak}\right)$$

$$\therefore y(t) = \frac{1}{a}\left(1 - e^{-ak}\right) u(t)$$

(Ans)





b) Given,

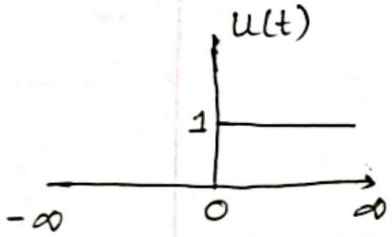
$$\begin{cases} x(t) = u(t) \\ x(k-t) = u(k-t) \end{cases}$$

$$y(t) = h(t) * x(t)$$

$$= \int_{-\infty}^{\infty} h(t) \cdot x(k-t)\, dt$$

$$= \int_{-\infty}^{\infty} e^{-at} u(t) \cdot u(k-t)\, dt$$

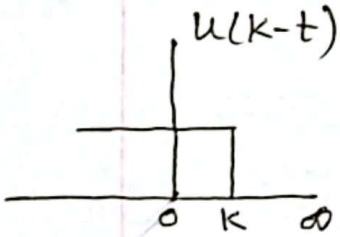
$$= \int_{-\infty}^{\infty} e^{-at} u(t) \cdot u(k-t)\, dt +$$



$$\int_{0}^{\infty} e^{-at} u(t)\, u(k-t)\, dt$$

$$= 0 + \int_{0}^{\infty} e^{-at} u(t)\, u(k-t)\, dt$$

$$= \int_{0}^{k} e^{-at} u(k-t)\, dt +$$



$$\int_{k}^{\infty} e^{-at} u(k-t)\, dt$$

$$= \int_{0}^{k} e^{-at}\, dt + 0$$

$$= \left[\frac{e^{-at}}{-a} \right]_{0}^{k}$$

$$= -\frac{1}{a}\left(e^{-ak} - e^{0}\right)$$

$$= -\frac{1}{a}\left(e^{-ak} - 1\right)$$

$$= \frac{1}{a}\left(1 - e^{-ak}\right)$$

$$\therefore y(t) = \frac{1}{a}\left(1 - e^{-ak}\right) u(t)$$

(Ans)

2.5

Given system,

$$h(t) = e^{-at}\, u(t)$$

$$x(t) = e^{at}\, u(-t)$$

Solve $y(t) = x(t) * h(t)$

Ans

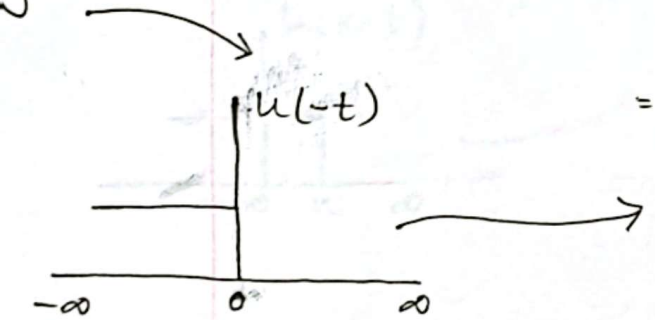
Given,

$$y(t) = x(t) * h(t)$$

$$= \int_{-\infty}^{\infty} x(t) \cdot h(k-t)\, dt$$

$$= \int_{-\infty}^{\infty} e^{at} \cdot u(-t) \cdot e^{-a(k-t)} \cdot u(k-t)\, dt$$

Unit step rotate



$$= \int_{-\infty}^{0} e^{at}\, u(-t) \cdot e^{-a(k-t)} \cdot u(k-t)\, dt$$

$$+ \int_{0}^{\infty} e^{at}\, u(-t) \cdot e^{-a(k-t)}\, u(k-t)\, dt.$$

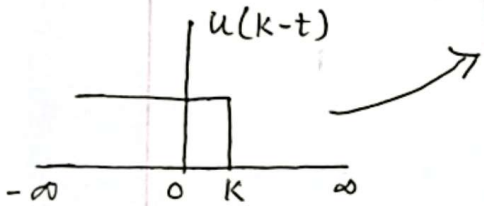
$$= \int_{-\infty}^{0} e^{at}\, e^{-ak+at}\, u(k-t)\, dt$$

$$+ 0$$

$$= \int_{-\infty}^{0} e^{at} \cdot e^{-ak} \cdot e^{at}\, u(k-t)\, dt$$

$$= \int_{-\infty}^{0} e^{2at} \cdot e^{-ak}\, u(k-t)\, dt$$

$$= \int_{-\infty}^{\infty} e^{2at} \cdot e^{-ak}\, u(k-t)\, dt + \int_{0}^{k} e^{2at} e^{-ak}\, u(k-t)\, dt$$



$$= \int_{-\infty}^{0} e^{2at} e^{-ak}\, dt + 0$$

$$= e^{-ak} \left[\frac{e^{2at}}{2a} \right]_{-\infty}^{0}$$

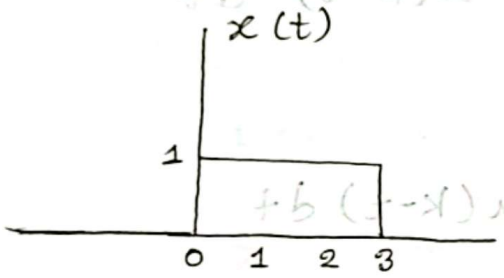
$$= e^{-ak} \left[\frac{e^{2a \cdot 0}}{2a} - \frac{0}{2a} \right]$$

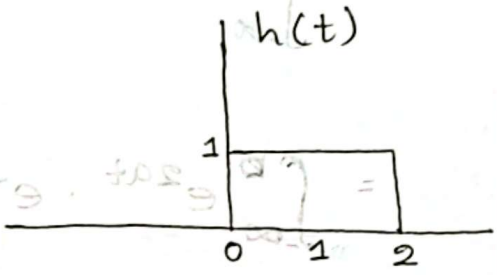
$$= e^{-ak} \cdot \frac{1}{2a} \cdot (1-0)$$

$$= \frac{e^{-ak}}{2a}$$

$$\therefore y(t) = \frac{e^{-ak}}{2a}\, u(t) \quad \text{(Ans)}$$

2.6



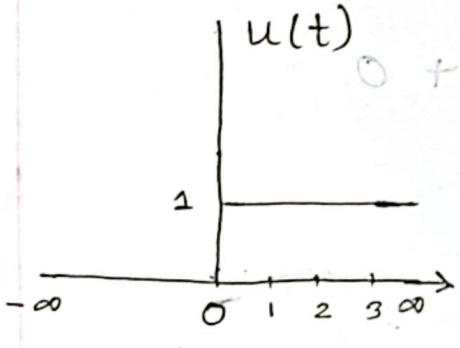


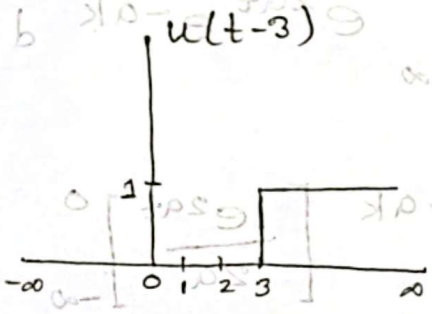
Evaluate $y(t) = x(t) * h(t)$. a) Mathematically

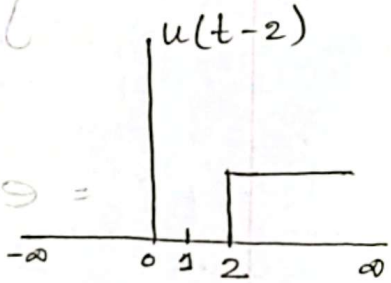
b) Graphically

Ans

Details for the graph →







So from graph,

$$x(t) = u(t) - u(t-3)$$

$$h(t) = u(t) - u(t-2)$$

Main ans-

a) From graph,

$$x(t) = u(t) - u(t-3)$$

$$h(t) = u(t) - u(t-2)$$

$\therefore y(t) = x(t) * h(t)$

$$= \int_{-\infty}^{\infty} x(t) \cdot h(k-t)\, dt$$

$$= \int_{-\infty}^{\infty} \{u(t) - u(t-3)\}\{u(k-t) - u(k-t-2)\}\, dt$$

$$= \int_{-\infty}^{\infty} u(t) \cdot u(k-t)\, dt - \int_{-\infty}^{\infty} u(t) \cdot u(k-t-2)\, dt$$

$$- \int_{-\infty}^{\infty} u(t-3) \cdot u(k-t)\, dt + \int_{-\infty}^{\infty} u(t-3) \cdot u(k-t-2)\, dt$$

So, let,

$$\int_{-\infty}^{\infty} u(t) \cdot u(k-t)\, dt \quad - - - - \text{①}$$

$$\int_{-\infty}^{\infty} u(t) \cdot u(k-t-2)\, dt \quad - - - \text{②}$$

$$\int_{-\infty}^{\infty} u(t-3) \cdot u(k-t)\, dt \quad - - - \text{③}$$

$$\int_{-\infty}^{\infty} u(t-3) \cdot u(k-t-2)\, dt \quad - - - \quad ④$$

From ①,

$$\int_{-\infty}^{\infty} u(t) \cdot u(k-t)\, dt$$

$$= \int_{-\infty}^{0} u(t) \cdot u(k-t)\, dt + \int_{0}^{\infty} u(t) \cdot u(k-t)\, dt$$

$$= 0 + \int_{0}^{\infty} u(k-t)\, dt$$

$$= \int_{0}^{k} u(k-t)\, dt + \int_{k}^{0} u(k-t)\, dt$$

$$= \int_{0}^{k} 1\, dt + 0$$

$$= [t]_0^k$$

$$= k - 0$$

$$= k$$

$$= k\, u(t)$$

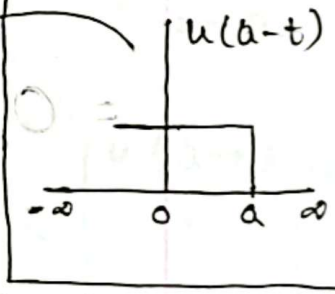
From ②,

$$\int_{-\infty}^{\infty} u(t)\, u(k-t-2)\, dt$$

$$= \int_{-\infty}^{0} u(t)\, u(k-2-t)\, dt + \int_{0}^{\infty} u(t)\, u(k-2-t)\, dt$$

$$= 0 + \int_{0}^{\infty} 1 \cdot u(a-t)\, dt$$

let, $(k-2 = a)$



$$= \int_{0}^{a} u(a-t)\, dt + \int_{a}^{\infty} u(a-t)\, dt$$

$$= \int_{0}^{a} 1\, dt$$

$$= [t]_{0}^{a}$$

$$= [t]_{0}^{k-2}$$

$$= k-2-0$$

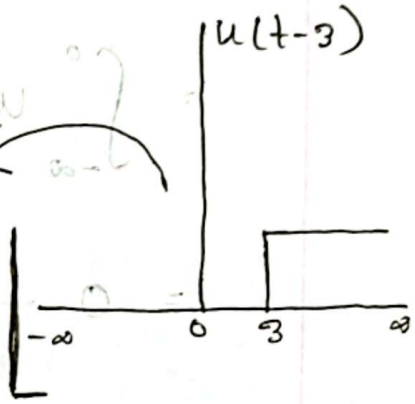
$$= k-2$$

$$= (k-2)\, u(t-2)$$

From ③,

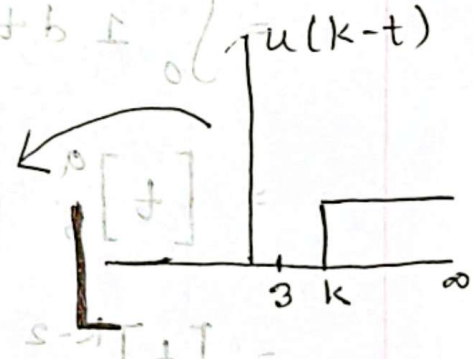
$$\int_{-\infty}^{\infty} u(t-3)\, u(k-t)\, dt$$

$$= \int_{-\infty}^{3} u(t-3) \cdot u(k-t)\, dt + \int_{3}^{\infty} u(t-3) \cdot u(k-t)\, dt$$



$$= 0 + \int_{3}^{\infty} 1 \cdot u(k-t)\, dt$$

$$= \int_{3}^{k} u(k-t)\, dt + \int_{k}^{\infty} u(k-t)\, dt$$



$$= \int_{3}^{k} 1\, dt$$

$$= [t]_{3}^{k}$$

$$= k - 3$$

$$= (k-3)\, u(t-3)$$

From ④,

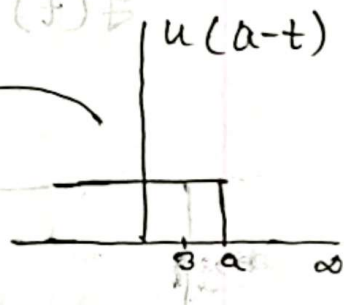
$$\int_{-\infty}^{\infty} u(t-3) \cdot u(k-t-2)\, dt$$

$$= \int_{-\infty}^{3} u(t-3) \cdot u(k-t-2)\, dt + \int_{3}^{\infty} u(t-3) \cdot u(k-t-2)\, dt$$

$$= 0 + \int_{3}^{\infty} u(k-2-t)\, dt$$

let, $k-2 = a$

$$= \int_{3}^{\infty} u(a-t)\, dt$$



$$= \int_{3}^{a} u(a-t)\, dt + \int_{a}^{\infty} u(a-t)\, dt$$

$$= \left[t\right]_{3}^{a} = a - 3$$

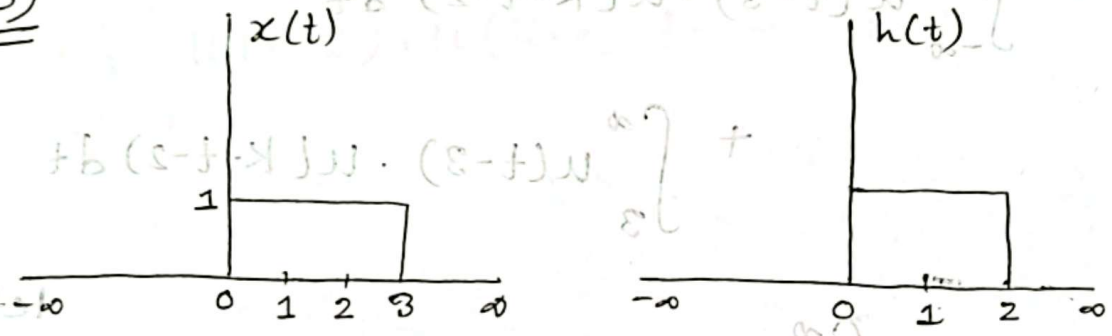
$$= k - 2 - 3$$

$$= k - 5$$

$$= (k-5)\, u(t-5)$$

$$\therefore y(t) = k\,u(t) + -(k-2)\,u(t-2) - (k-3)\,u(t-3) + (k+5)\,u(t-5)$$

b)



$$y(t) = x(t) * h(t)$$

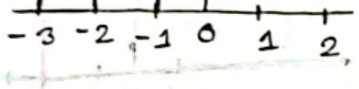
Rule

Folding → Shifting → Multiplication

Next page —>

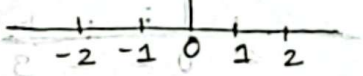
h(k-t)

$t < -2$



x(t) · h(k-t)

$t < 0$

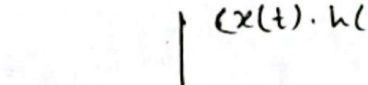


h(k-t)

$-2 \leq t \leq 0$

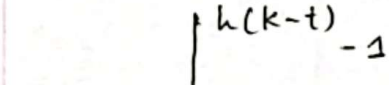


x(t) · h(k-t); $-2 \leq t < 0$

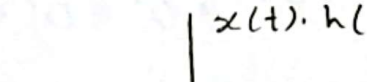


h(k-t)

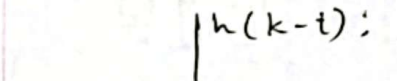
$-1 \leq t \leq 1$



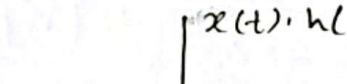
x(t) · h(k-t);



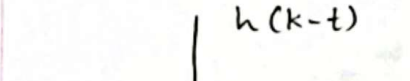
h(k-t);



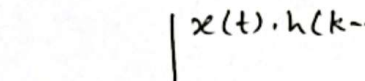
x(t) · h(k-t)



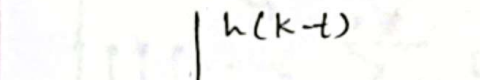
h(k-t)



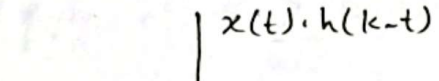
x(t) · h(k-t)



h(k-t)



x(t) · h(k-t)

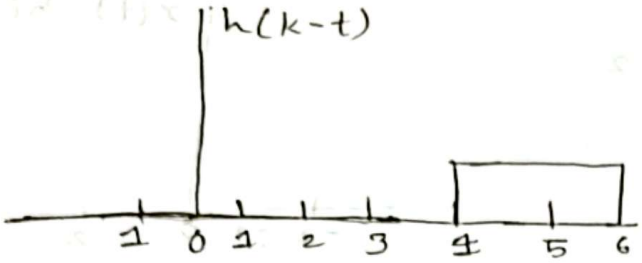


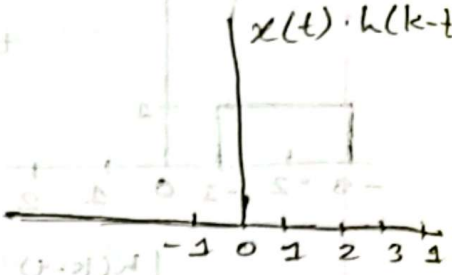


x(t) · h(k-t)



P.T.O


h(k-t)
-1
0
1
2
3
4
5
6


x(t)·h(k-t)
-1
0
1
2
3
4

2.28

Given system, $x[n] = u[n]$

$$h[n] = a^n u[n]$$

a) $y[n] = x[n] * h[n]$

b) $y[n] = h[n] * x[n]$

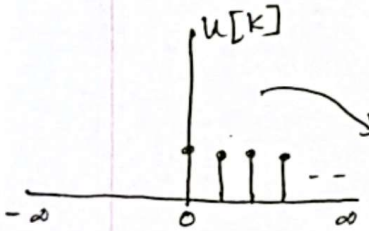
Ans

a)

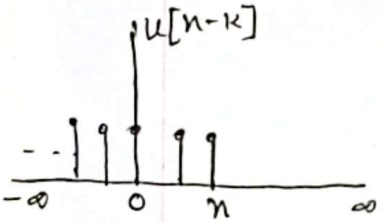
$$y[n] = x[n] * h[n]$$

$$= \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k]$$

$$= \sum_{k=-\infty}^{\infty} u[k] \cdot a^{n-k} u[n-k]$$



$$= \sum_{k=-\infty}^{\infty} u[k] \cdot a^{n-k} \cdot [u[n-k] + \sum_{k=0}^{\infty} u[k] \cdot a^{n-k} u[n-k]$$



$$= 0 + \sum_{k=0}^{\infty} a^{n-k} u[n-k]$$

$$= \sum_{k=0}^{n} a^{n-k} u[n-k] + \sum_{k=n}^{\infty} a^{n-k} u[n-k]$$

$$= \sum_{k=0}^{n} a^{n-k}$$

$$= \sum_{p=n}^{0} a^{p}$$

$$= \sum_{p=0}^{n} a^{p}$$

$$= a^0 + a^1 + a^2 + a^3 + \text{- - - - -} + a^n$$

$$= \frac{1 - a^{n+1}}{1-a}$$

$$= \frac{1 - a^{n+1}}{1-a}\, u[n]$$

let,

$k=0$	$k=n$
$n-k=p$	$n-n=p$
$\Rightarrow n-0=p$	$p=0$
$\Rightarrow p=n$	
lower limit	Upper limit

$\frac{a^1}{1} = a$

$\frac{a^2}{a} = a \; ; r = a$

$\frac{a^3}{a} = a$

গুণোত্তর ধারা,

n-তম পদের সমষ্টি $= \frac{1-r^{n+1}}{1-r}$

b)

$$y[n] = h[n] * x[n]$$

$$= \sum_{k=-\infty}^{\infty} h[k] \cdot x[n-k]$$

$$= \sum_{k=-\infty}^{\infty} b \cdot a^k u[k] \cdot u[n-k]$$

$$= \sum_{k=0}^{\infty} a^k u[n-k]$$

$$= \sum_{k=0}^{n} a^k$$

$$= a^0 + a^1 + a^2 + \cdots + a^n$$

গুণোত্তর ধারার সমষ্টি,

$$= \frac{1 - a^{n+1}}{1-a}$$

2.29

Compute $y[n] = x[n] * h[n]$

a) $x[n] = \alpha^n u[n]$; $h[n] = \beta^n u[n]$

b) $x[n] = \alpha^n u[n]$; $h[n] = \alpha^{-n} u[-n]$

Ans

a)

$$y[n] = x[n] * h[n]$$

$$= \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k]$$

$$= \sum_{k=-\infty}^{\infty} \alpha^k u[k] \cdot \beta^{n-k} u[n-k]$$

$$= \sum_{k=0}^{\infty} \alpha^k \beta^{n-k} u[n-k]$$

$$= \sum_{k=0}^{n} \alpha^k \beta^{n-k}$$

$$= \sum_{k=0}^{n} \alpha^k \cdot \beta^n / \beta^k$$

$$= \beta^n \sum_{k=0}^{n} \left(\frac{\alpha}{\beta}\right)^k$$

$$= \beta^n \cdot \frac{1 - \left(\frac{\alpha}{\beta}\right)^{n+1}}{1 - \frac{\alpha}{\beta}}$$

If,
$$\alpha \neq \beta$$

$$\therefore y[n] = \beta^n \cdot \frac{1 - \left(\frac{\alpha}{\beta}\right)^{n+1}}{1 - \frac{\alpha}{\beta}} = \beta^n \frac{1 - \left(\frac{\alpha}{\beta}\right)^{n+1}}{1 - \frac{\alpha}{\beta}} u[n]$$

If,
$$\alpha = \beta$$

$$\therefore y[n] = \alpha^n (n+1) = \alpha^n (n+1)\, u[n]$$

$$\therefore y[n] = \begin{cases} \beta^n \dfrac{1 - \left(\frac{\alpha}{\beta}\right)^{n+1}}{1 - \frac{\alpha}{\beta}} u[n], & \alpha \neq \beta \\ \alpha^n (n+1), & \alpha = \beta \end{cases}$$

Ans

b)

$$y[n] = x[n] * h[n]$$

$$= \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k]$$

$$= \sum_{k=-\infty}^{\infty} \alpha^k u[k] \cdot \alpha^{-n+k} u[-n+k]$$

$$= \sum_{k=0}^{\infty} \alpha^k \cdot \alpha^{-n} \cdot \alpha^k \; u[-n+k]$$

$$= \alpha^{-n} \sum_{k=0}^{n} \alpha^{2k}$$

$$= \alpha^{-n} \frac{1-(\alpha^2)^{n+1}}{1-\alpha^2}$$

$$= \alpha^{-n} \frac{1}{1-\alpha^2} \quad \left(\text{Geometric series, } \frac{1}{1-r} \text{ when, } |r| < 1\right)$$

$$= \frac{\alpha^{-n}}{1-\alpha^2} \; u[n]$$

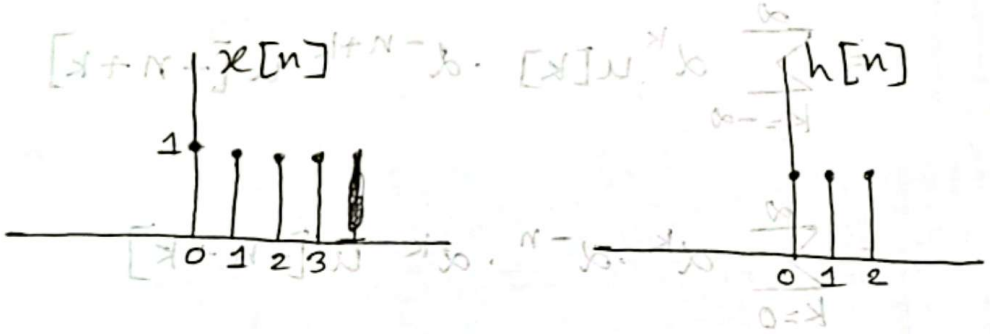
(Ans)

2.30

Evaluate, $y[n] = x[n] * h[n]$

a) Analytically

b) Graphically,



Ans

a)

$$x[n] = \delta[n] + \delta[n-1] + \delta[n-2] + \delta[n-3]$$

$$h[n] = \delta[n] + \delta[n-1] + \delta[n-2]$$

$$\therefore y[n] = x[n] * h[n]$$

$$= x[n] * \{\delta[n] + \delta[n-1] + \delta[n-2]\}$$

$$= x[n] * \delta[n] + x[n] * \delta[n-1] + x[n] * \delta[n-2]$$

$$= x[n] + x[n-1] + x[n-2]$$

P.T.O

$$= \delta[n] + \delta[n-1] + \delta[n-2] + \delta[n-3] + \delta[n-1]$$

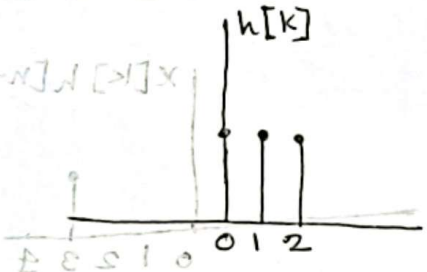
$$+ \delta[n-2] + \delta[n-3] + \delta[n-4] + \delta[n-2] +$$

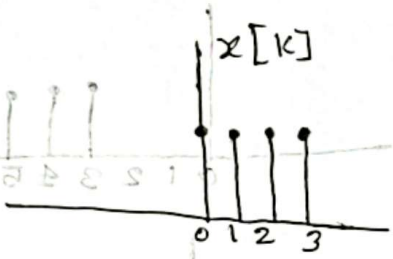
$$\delta[n-3] + \delta[n-4] + \delta[n-5]$$

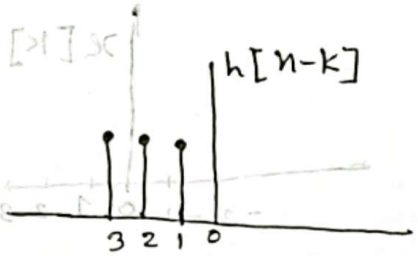
$$= \delta[n] + 2\delta[n-1] + 3\delta[n-2] + 3\delta[n-3] + 2\delta[n-4]$$

$$+ \delta[n-5]$$

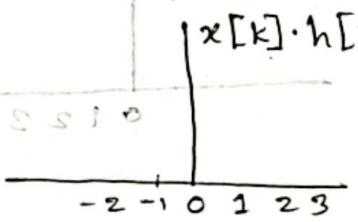
b) Graphical →

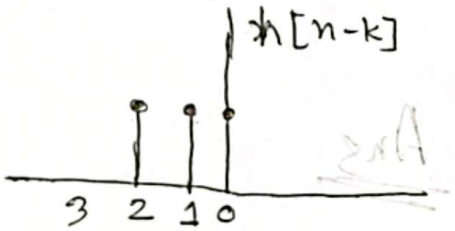




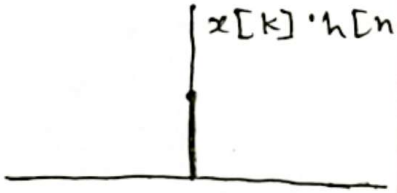


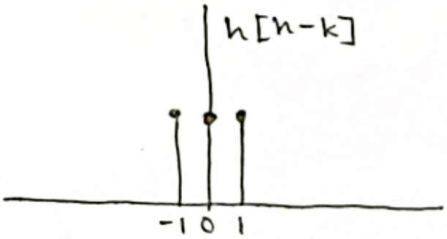
$n<0$



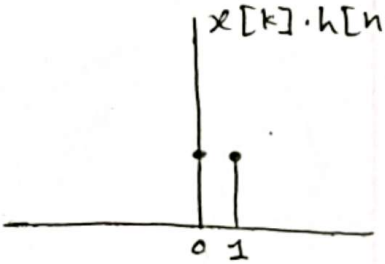


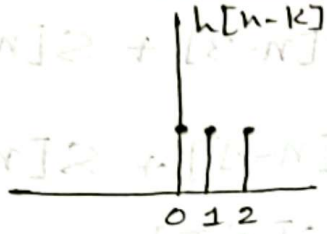
$n=0$



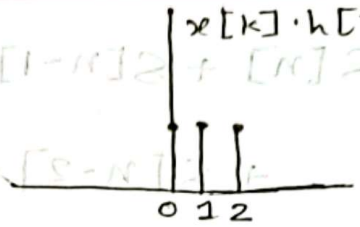


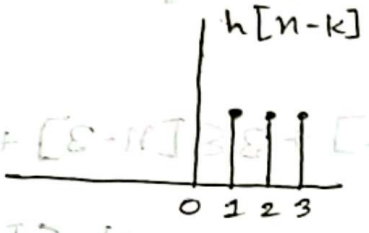
$n=1$



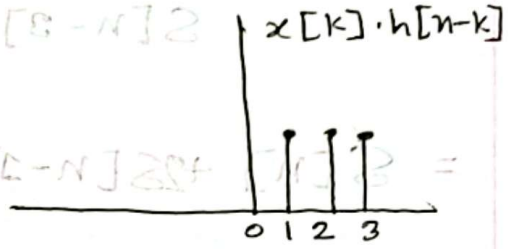


$n = 2$



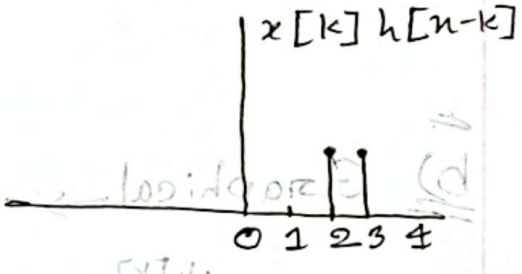


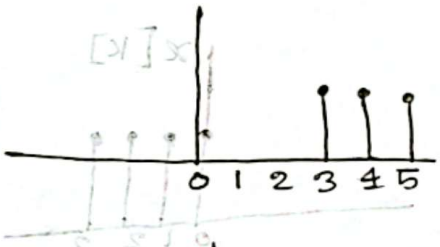
$n = 3$





$n = 4$



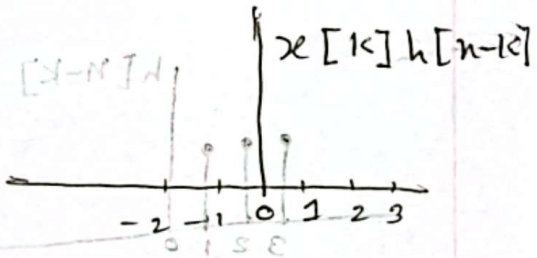


$n = 5$





$n > 5$



Ans